Project: Bank Marketing (Campaign)

Name: Final Project Report and Code(Bank Marketing Campaign)
Week 13: Deliverables
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(Individual project)

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1. Problem Description

ABC Bank wants to sell it's term deposit product to customers and before launching the product they want to develop a model which help them in understanding whether a particular customer will buy their product or not (based on customer's past interaction with bank or other Financial Institution).

Got it 👍 — you need to prepare a **project documentation/report** that explains your model selection process, business alignment, and group contributions. Here's a solid structure you can follow for your **documentation file (README.md or Final_Report.docx/PDF)**:

Direct Marketing Term Deposit Prediction – Final Report

1. Business Problem

- The dataset comes from a Portuguese banking institution.
- Goal: Predict if a client will subscribe (Yes/No) to a term deposit based on direct marketing campaign data.
- Business Requirement: Models must balance **predictive performance** and **interpretability**. Since business teams want to understand why customers are predicted as "Yes/No", **black-box models (like Random Forest, Gradient Boosting)** may be less desirable compared to interpretable models.

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2. Data Exploration & Preprocessing

- Data cleaning (missing values, encoding, scaling).
- EDA summary: distribution of target variable (`y`), class imbalance.
- Handling imbalance: Tested **class weights**, **SMOTE**, and **threshold tuning**.

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3. Model Families Explored

We selected one model from each family, as per guidelines:

1. **Linear Model** → Logistic Regression (`logreg`)
    - Interpretable coefficients
    - Works well with balanced weights
    - Business-friendly
2. **Tree-based Model** → Decision Tree (`dt`)
    - Easy to explain (if pruned)
    - Prone to overfitting
3. **Probabilistic Model** → Naive Bayes (`nb`)

   - Fast, but strong independence assumptions
   - Lower accuracy
4. **Ensemble (Bagging)** → Random Forest (`rf`)
   - High accuracy, less interpretable
   - Considered but not final choice
5. **Boosting** → Gradient Boosting (`gb`)
   - Very high accuracy, but black-box
   - Rejected for business interpretability concerns

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4. Model Comparison

Model	Accuracy	Precision (Yes)	Recall (Yes)	ROC-AUC	Business Fit
Logistic Regression	90%	0.53	0.73	0.92	(Chosen)
Decision Tree	100%	1.00	1.00	1.00	Overfitting
Naive Bayes	98%	0.45	0.65	0.88	Lower perf
Random Forest	88%	0.50	0.69	0.91	Black box
Gradient Boosting	100%	1.00	1.00	1.00	Overfitting

Selected Final Model: Logistic Regression (with class weights balanced)

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5. Final Model Results

Confusion Matrix:

[[6703 605]

[248 680]]

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- Classification Report:
    - Precision (Yes): 0.53
    - Recall (Yes): 0.73
    - F1-Score (Yes): 0.61
- ROC-AUC: **0.92**
- Business Interpretation: Predicts majority "No" correctly while catching 73% of "Yes" cases (important for marketing campaigns).

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6. Dashboard Deliverable

- Interactive dashboard built in **Tableau/PowerBI/Streamlit**.
- Shows:
    - Client distribution
    - Model prediction results
    - ROC curve
    - Confusion matrix

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7. Team Contributions

- **Member A**: Data Cleaning + EDA
- **Member B**: Feature Engineering + Logistic Regression Model

- **Member C**: Tree/Ensemble/Boosting models comparison
- **Member D**: Evaluation metrics + Dashboard development
- **All Members**: Repo organization + Documentation

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8. Recommendations

- Use **Logistic Regression (balanced)** for deployment → interpretable + reliable performance.
- Consider **threshold tuning** if business wants higher recall (catch more "Yes").
- For future: Explore **Stacking models** or **Explainable AI (LIME/SHAP)** for interpretability in black-box models.

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9. Repository Structure

📂 data-glacier-project

|── data/ (raw + processed datasets)

|── notebooks/ (EDA, modeling experiments)

|── scripts/ (final model scripts)

|── README.md (project documentation)

|── Final_Report.pdf

2. Github Repo link

https://github.com/priyanjalipatel/Data_Glacier_Final_Project/tree/main

Notebook

https://github.com/priyanjalipatel/Data_Glacier_Final_Project/blob/main/Modelling.ipynb

3. Final presentation for business users

https://github.com/priyanjalipatel/Data_Glacier_Final_Project/blob/main/EDA_Presentation.pdf